

SURENDRANATH COLLEGE
Mode of Examination: Online
M.Sc. (Computer Science)
Semester-II Intermediate Examination, 2020
Subject: Computer Science
Paper: CSM203 (Object Oriented Systems)
Full Marks: 50
Time: 2:30 Hours
Date: 11/12/2020

Duration: 11:00 hrs. – 13:30 hrs.

Please follow the following instructions carefully.

The figures in the margin indicate full marks.

Candidates are required to answer in their own words as far as applicable.

Each Page of answer scripts should have your examination Roll Number on the right side top corner of your answer script and the page number on bottom.

The name of the scanned copy of the answer script will be of the following format:

MSC_CSM201_Roll-Number.pdf

(Example: MSC_CSM201_C91/CSC/181001)

The subject of the mail should be answer-script file name.

The scanned answer scripts are to be sent to- **snccsrc@gmail.com**

No answer script is valid if received after 13.30 hrs. on the Date of Examination. Write the answers with black/blue ink ball pen.

SECTION – A

F.M-20

Group – A

5x2=10

(Any Five)

1. Answer the following:

- a. What is Creational Design Pattern?
- b. Define MVC design Pattern.
- c. Explain the applicability of Factory method.
- d. When we can use Prototype design pattern?
- e. Write down the drawbacks of Flyweight design pattern.
- f. How we can relate Chain of Responsibility design pattern with Command design pattern
- g. Define GUI Programming.
- h. How we can read files from URL using JAVA?

Group-B

2x5=10

(Any Two)

2. Answer the following:

- a. Define Creational design patterns. How the Flyweight design pattern can be related with Singleton design pattern? [2+3]
- b. Define Behavioral design patterns. Discuss the advantages and disadvantages of Strategy design pattern. [2+3]
- c. What is Matchers? Explain the different types of matchers. [2+3]
- d. Define applet. Draw and explain the lifecycle of an applet. [2+3]
- e. What are the key features of Model View Controller design pattern? Write down the differences between structural and behavioural design pattern. [2+3]

Section – B

F.M – 30

Group – A

5x2= 10

(Any *Five*)

1. Answer the following:
 - a. How we can restrict inheritance in a class?
 - b. What is Activity Diagram?
 - c. Write down the Advantages of Abstract Data Type.
 - d. Explain the function of the keyword “***final***”
 - e. Why Java doesn’t use pointers?
 - f. What is Method Overriding?
 - g. What do you mean by the term “***Platform independence***”
 - h. Differentiate between Array and Array List.

Group-B

4x5 = 20

(Any *Four*)

2. Answer the following:
 - a. How many types of exceptions are there? Explain. [2+3]
 - b. Why Java doesn’t support Multiple Inheritance? [5]
 - c. What is the difference between this() and super() in Java? What is Interface in java? [3+2]
 - d. Write a short note on In memory architecture of JRE. [5]
 - e. Write the use of the keyword “***Finally***”. What is the motivation behind Collection framework? [2+3]
 - f. Which data structure is used in Garbage Collector? Explain why? [1+4]
 - g. How is an Abstract class different from an Interface? [5]